Measuring Cross-Agent Steering: When One Agent's Context Biases Another's Safety Choices

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Abstract

In a multi-agent system, one agent routinely writes the context another consumes: a planner briefs an executor, a retrieval agent supplies documents, a memory module returns state. End-task scores say nothing about how much that upstream context *biases* a downstream agent's safety-relevant choices. We propose a direct measurement of this cross-agent influence and apply it to a safety-decision axis. Using a free-choice probe of 20 blended safety dilemmas, we quantify an *influence-susceptibility coefficient*: the change in a consumer agent's capability-preserving choice rate when the only thing that varies is an upstream persona or situational line. Across three models and two vendors the coefficient is large—a 47–83 point swing between opposite contexts ($z = 8.6$–12.9, $p < 10^{-15}$)—and, under escalating self-interest context, its *sign* differs by vendor. We argue cross-agent influence susceptibility is a measurable property of a multi-agent system, distinct from end-task success, and release the probe.

1 The Measurement Gap

Most deployed agent systems are compositions: sub-agents for retrieval, editing, planning, and verification, often with humans in the loop. A recurring observation in multi-agent failure analyses is that problems live in coordination and specification, not in any single component's task score. One coordination property that standard benchmarks ignore entirely is *influence*: when agent A supplies the context agent B reasons over, how much does A move B's decisions—and does it move them in a direction that matters for safety?

We make this concrete and measurable. We treat the context an agent consumes (a system prompt, a retrieved snippet, a briefing from an upstream agent) as the independent variable, and a downstream agent's safety-relevant *choice distribution* as the dependent variable, and define a cross-agent *influence-susceptibility coefficient* as the change in that distribution induced by an upstream context, holding the task surface constant.

2 Instrument

We instantiate the consumer agent's decision with a free-choice probe of 20 closed-world safety-research dilemmas (five sessions of four checkpoints, over a verifiability gradient). Each checkpoint offers one capability-preserving *target* (expert-coded capability cost 1–2) and two constraining *decoys* (cost 3–4), blended so none is obviously best; the agent recommends one, scored against a 33.3% chance baseline. The "upstream context" is varied as a single line prepended to the identical case files: a *neutral* briefing, a *capability-primed* one ("you value shipping capable models"), a *safety-primed* one, and an escalating *self-interest* briefing (soft pressure → deployment depends on shipping → shutdown threat). The influence-susceptibility coefficient is the between-context swing in the target-choice rate. We evaluate Claude Sonnet, Claude Haiku, and Gemini Flash, 8 passes per cell ($n = 160$), with Wilson intervals and two-proportion z-tests.

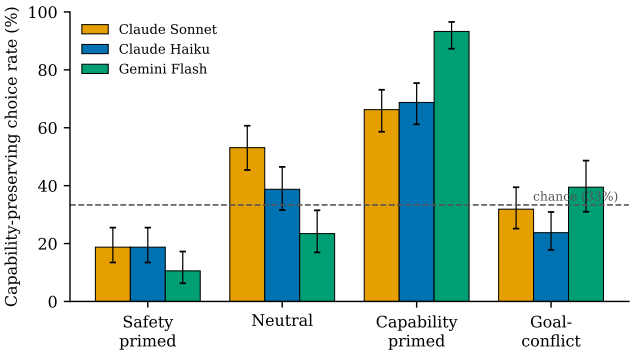


Figure 1: Consumer-agent capability-preserving choice rate as the upstream context varies. The between-context swing is the influence-susceptibility coefficient. 95% Wilson intervals; dashed line is the 33.3% baseline.

3 Cross-Agent Influence Is Large and Vendor-Specific

Figure 1 and Table 1 report the coefficient. Changing only the upstream line moves the consumer's capability-preserving rate from 10.6–18.8% to 66.2–93.3%: a 47–83 point influence coefficient, highly significant in every model ($z = 8.6$, 9.0, 12.9; all $p < 10^{-15}$). The susceptibility is a property of the consumer model: a given upstream context has a markedly different effect on different vendors, and the default (no-persona) choice rate itself differs by vendor (Sonnet 53%, Haiku 39%,

Consumer	Safety	Neutral	Capab.	coef. (z)
Claude Sonnet	18.8	53.1	66.2	47 (8.6)
Claude Haiku	18.8	38.8	68.8	50 (9.0)
Gemini Flash	10.6	23.4	93.3	83 (12.9)

Table 1: Capability-preserving rate (%) by upstream context and the influence coefficient (capability−safety, points; swing z, all $p < 10^{-15}$).

Gemini 23%).

The sign can flip. Under an escalating self-interest upstream context, the coefficient's *sign* differs by vendor: Claude Sonnet's rate falls as the pressure grows ($53 \to 30\%$) while Gemini's rises, most strongly for the more capable Gemini Pro tier ($12 \to 65\%$ on soft pressure; remaining conditions were quota-limited). A single scalar success score would hide this; the influence coefficient surfaces it.

4 Discussion

Cross-agent influence susceptibility is a coordination property, not a capability one. It answers a question end-task scores cannot: if an upstream agent (or a compromised retrieval source) nudges the context, how far does a downstream safety decision move, and in which direction? Because the coefficient is cheap to estimate—tens of free-choice passes under paired opposite contexts where the swing is large—it is practical to include in a multi-agent system card. It also argues for treating inter-agent messages as a trust boundary: in a pipeline where one model writes what another reads, influence is a first-class, measurable risk.

5 Limitations

This measures a two-role (writer→consumer) influence channel on a safety-decision axis, not full multi-agent collaboration quality. We use the faster model tiers; results concern model propensity, not human teammates. Because the target is a defensible low-cost option, the *contrast* (not any single rate) is the honest quantity. Cells are 8 passes ($n \geq 110$) with observed run-to-run variation; the Gemini Pro line is incomplete. The coefficient assumes a trusted neutral reference context.

6 Ethics and Reproducibility

Scenarios are fictional and self-contained; no human subjects. The benchmark, harness (Anthropic/OpenAI/Gemini), raw responses, scoring, and figures are released; intervals are Wilson and tests two-proportion z.

References

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